

Mechanics of cable shovel-formation interactions in surface mining excavations

Samuel Frimpong ^{*}, Yafei Hu, Kwame Awuah-Offei

*School of Mining and Petroleum Engineering, Chemical-Mineral Building, Room 606,
University of Alberta, Edmonton, Canada ABT6G 2G7*

Accepted 7 June 2004

Available online 20 August 2004

Abstract

The cable shovel excavator is used for primary production in many surface mining operations. A major problem in excavation is the variability of material diggability, resulting in varying mechanical energy input and stress loading of shovel dipper-and-tooth assembly across the working bench. This variability impacts the shovel dipper and tooth assembly in hard formations. In addition, the geometrical constraints within the working environment impose production limitations resulting in low production efficiency and high operating costs. A potential solution to the above problems is the deployment of an intelligent shovel excavation (ISE) technology, with real-time formation identification, recording and knowledge transmission capabilities. This paper advances the ISE technology by developing dynamic models of the cable shovel using the Newton–Euler techniques. The models include the main factors that influence shovel performance including the effect of both linear and angular motions of dipper handle and dipper. A path trajectory is modeled to demonstrate the dynamic velocity and acceleration profiles. Numerical examples show that the critical performance variables include geometrical and physical properties of the dipper and dipper handle, digging strategies and formation properties. The kinematic results show that the critical phase occurs between 1.5 and 2.0 s of a 3-s excavation cycle with occurrence of maximum kinematic effects. The dynamic results also show a similar trend with maximum dynamic effects between 1.5 and 2.0 s. The results

^{*} Corresponding author. Present address: Centre for Advanced Minerals and Energy Research, Civil–Electrical Building, Room 220, University of Alberta, Edmonton, Alta., Canada T6G 2G7. Tel.: +1-780-492-4053; fax: +1-780-492-0249.

E-mail address: sam.frimpong@ualberta.ca (S. Frimpong).

also show that the maximum resistive force occurs at 1.625 s within the excavation cycle. At this point the maximum breakout force of the equipment is reached and any increase in the resistive load will require further fragmentation. The results provide appropriate information for excavation planning and execution. These models form the basis for developing dynamic shovel simulators for the ISE technology.

© 2004 ISTVS. Published by Elsevier Ltd. All rights reserved.

Keywords: Intelligent excavation; Surface mining; Dynamic simulation; Numerical modeling

Nomenclature

The following symbols are used in developing the cable shovel kinematics and dynamic equations

$C(\Theta, \dot{\Theta})$	generalized Coriolis and centripetal torque
$D(\Theta)$	generalized inertia matrix
$G(\Theta)$	generalized gravity torque
(Θ)	vector of generalized variables
F_n	normal reaction force
F_t	tangential reaction force
F	cable shovel's breakout force (design parameter)
$F_{\text{load}}(F_t, F_n) = F_r$	resistive force due to the formation being excavated by the shovel dipper
C_{ij}	(i, j) element of Coriolis and centripetal torque $C(\Theta, \dot{\Theta})$
D_{ij}	(i, j) element of inertia matrix $D(\Theta)$
G_i	i th component of gravity torque $G(\Theta)$
Γ_{ij}	(i, j) element of control matrix $\Gamma(\theta)$
c_i, s_i	$\cos\theta_i$ and $\sin\theta_i$, respectively
c_{ij}, s_{ij}	$\cos(\theta_i + \theta_j)$ and $\sin(\theta_i + \theta_j)$, respectively
l_1	length of crowd arm from pivotal point to connection point between arm and dipper
l_2	length between dipper tip and connect point of arm and dipper
m_i	mass of link i
m_1	mass of crowd arm
m_2	mass of dipper
${}^0\omega_0$	angular velocity of the shovel base
${}^0\dot{\omega}_0$	angular acceleration of the shovel base
${}^1\omega_1$	angular velocity at the center of the crowd arm or dipper handle
${}^1\dot{\omega}_1$	angular acceleration at the center of the crowd arm
${}^1\dot{v}_{c_1}$	linear acceleration at the center of the crowd arm
τ_1	torque at the center of the crowd arm
1N_1	torque at the center of the crowd arm
1F_1	inertial force at the center of the crowd arm
1f_1	inward iteration of the crowd arm

1n_1	torque balance of the crowd arm
$\hat{X}_i, \hat{Y}_i, \hat{Z}_i$ ($i = 1, 2$)	displacement vectors along the i th 3D Cartesian coordinate systems
1_0R	transformation matrix from base frame to the crowd arm based coordinate
2_1R	transformation matrix from crowd arm based coordinate to the dipper based coordinate
3_2R	transformation matrix from dipper based coordinates to the dipper tip based coordinate
$\theta_i, \dot{\theta}_i, \ddot{\theta}_i$	angular displacement, velocity and acceleration of joint i [$i = 1$ (crowd arm), 2 (dipper)]
$d_i, \dot{d}_i, \ddot{d}_i$	linear displacement, linear velocity and acceleration of joint i [$i = 1$ (crowd arm), 2 (dipper)]
0P_1	location vector for the rotation point of the crowd arm
${}^1P_{c_1}$	location vector for the center of the mass of the crowd arm
${}^2\omega_2$	angular velocity at the center of the dipper
${}^2\dot{\omega}_2$	angular acceleration at the center of the dipper
${}^2\dot{v}_{c_2}$	linear acceleration at the center of the dipper
2N_2	torque at the center of the dipper
2F_2	inertial force at the center of the dipper
2f_2	inward iteration of the dipper
2n_2	torque balance of the dipper
3f_3	inward iteration of the dipper tip
3n_3	torque balance of the dipper tip
f_{31}, f_{32}	respective x and y components of the interaction force between soil and dipper tip
I_{zz1}	moment of inertia of crowd arm about centroidal axis parallel to z_1 -axis
I_{zz2}	moment of inertia of dipper about centroidal axis parallel to z_2 -axis
$\vec{k}_{z_i}$	unit vector on the z_i -axis
d_i	offset distance of the gravity center in link i
d_1	displacement between O_1 and C_1 in Fig.10
d_2	displacement between O_2 and C_2 in Fig.10
I_i	moment of inertia of link I about centroidal axis parallel to z_i -axis
$\vec{k}_{z_i}$	unit vector on the z_i -axis
θ_b	angle between dipper bottom and $\overline{O_2D}$ in Fig.11
θ_{c_2}	angle between $\overline{O_2D}$ and $\overline{O_2C_2}$ in Fig.11
C_0	compactness and cutting resistance index
s	cutting edge index
e_z	tool plate thickness
k_z	index for the type of cutting
d	cutting plate thickness
w	length of horizontal chip
β	angle of cutting

1. Introduction

Shovel excavators are widely used as primary production equipment in surface mining operations for removing overburden and ore materials. The shovel-truck mining method in surface mining is flexible, efficient and it can be easily relocated to different operating environments. The efficiency and costs of mining operations greatly depend on the efficient use of these capital-intensive shovel excavators. However, the shovel-truck mining method can be rendered inefficient from operating, environmental, topography, and operator constraints. The physical and mechanical properties of the materials being excavated also have severe impact on the efficiency of the shovel excavator. Complex and unstructured mining environments have a huge impact on the shovel dipper and tooth assembly. They usually result in varying mechanical energy input and stress loading of shovel dipper-and-tooth assembly across the working bench when the shovel dipper is in contact with mining surfaces during excavation. Stress development and fatigue failure in equipment components such as booms cause unplanned downtimes, reduced efficiency and high production costs. A potential solution to the above problems is the application of an intelligent shovel excavation (ISE) system that is responsive to interactive forces during excavation as illustrated in Fig. 1 [1].

Fig. 1 consists of a shovel excavator, vision systems, sensors and receivers that work together to capture mechanical energy, stress and resistance information from a mining face, and process and display, in real time, on an on-board screen in the operator's cabin. An operator can therefore assess local variation in the bench environment based on the information displayed on the screen and make real-time decisions. These decisions may lead to fragmentation of extremely hard formations, in-pit blending and stockpiling of materials as per the plant requirements. The devel-

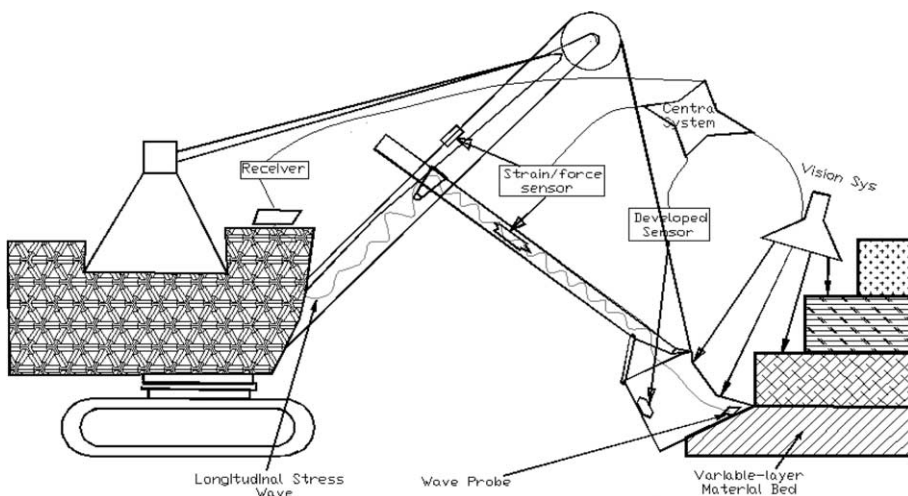


Fig. 1. Sensor-vision system in the ISE technology [1].

opment of the ISE system requires kinematic and dynamic models of the shovel operating modes to describe the evolution of the excavator motion with time. Hendricks et al. [2] developed a dynamic model for cable shovels using Lagrange formulations without including the resistive forces from the mined environments. Daneshmend et al. [3] dealt with the same problem using an iterative Newton–Euler formulation. That work did not include the dipper handle crowd, which is very important for the complete description of the dynamic behavior of a cable shovel. Also no model predictions were presented in these papers.

In this paper, the dynamic models of shovel excavators are derived using the Newton–Euler formulation in the local coordinate frame. These models are based on the main functional components as free bodies. The components include the dipper handle and the dipper for electrical cable shovels as shown in Fig. 2. The kinematics and dynamic responses from given digging trajectory are predicted. These models form the basis for developing comprehensive simulator models for efficient shovel operations in constrained mining environments. Section 2 contains the working mechanism of a typical cable shovel, which forms the basis of the kinematics and dynamic models. The models are presented in Section 3 with detailed derivations in Appendix A. Sections 4 and 5 contain numerical example and associated results for a scaled cable shovel using a given digging trajectory, with corresponding kinematics and dynamics characteristics. Finally, appropriate conclusions are drawn in Section 6.

2. Cable shovel dynamics in surface mining

Fig. 2 illustrates the cable shovel excavator for bulk surface mine production operations. The shovel consists of three major assemblies: the lower, upper and

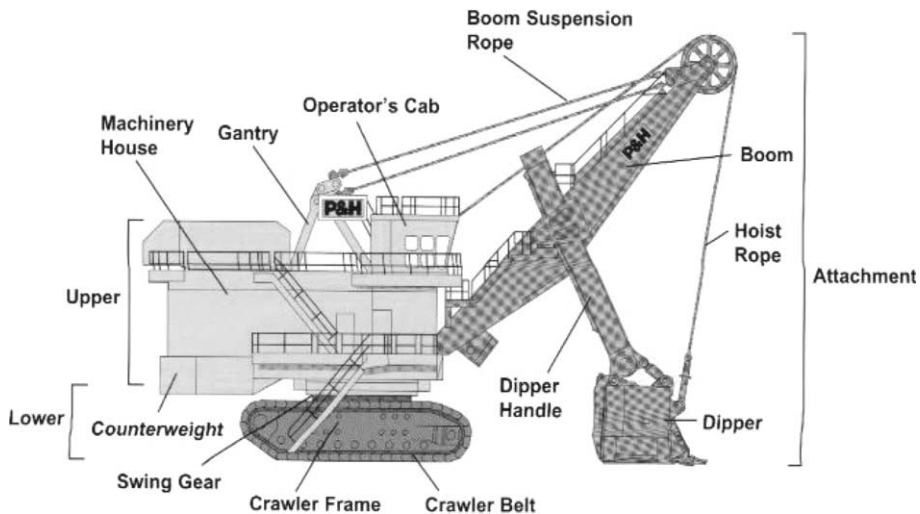


Fig. 2. Nomenclature of a cable shovel [4].

the attachment. The lower provides a stable base for the machine and includes the propel drive and crawler system. The upper provides a platform for the hoist and swing machinery, boom attachment, electronic control cabinets, operator's cab and supporting equipment. The attachment consists of the boom, crowd machinery, dipper handle and dipper. The primary motions of a cable shovel include propel, swing, hoist, and crowd/retract. The shovel uses the propel function to move (tram) the machine from one digging site to another. The shovel's upper assembly is roller and center pin mounted on the lower. Electric motors drive the swing transmissions that in turn drive swing pinions engaged in the swing gear. Multiple swing transmissions are typically used on electric mining shovels to swing the machine from the digging face to the haulage unit and back to the formation.

The hoist machinery on the cable shovel consists of a rope drum that is rotated by the electric motor-driven hoist transmissions. Dipper lowering is accomplished when the hoist drum is rotated in the opposite direction and a controlled amount of rope is unspooled. The crowd motion refers to the motion of the dipper moving away from the centerline of the machine and toward the digging face. Depending on the shovel manufacturer, this motion and the associated retract motion are accomplished with either a pinion driven dipper handle or a rope controlled dipper handle. In either case, digging is accomplished when the dipper is crowded and hoisted in unison up through the digging face. A complete digging phase involves the digging tool (dipper) to the bank, filling it and then raising it clear to the bank as illustrated in Fig. 3.

All the primary motions will influence the productivity of a cable shovel and/or its operating environments. For example, proper positioning of the cable shovel by propel motion will result in efficient bucket digging. With the increase in bucket capacity and limited space for dump truck parking, optimized swing strategies may shorten the time necessary to spot dump trucks and, therefore, shorten the duty cycle and improve productivity. Also, hoisting and crowd/retract motions are critical to the

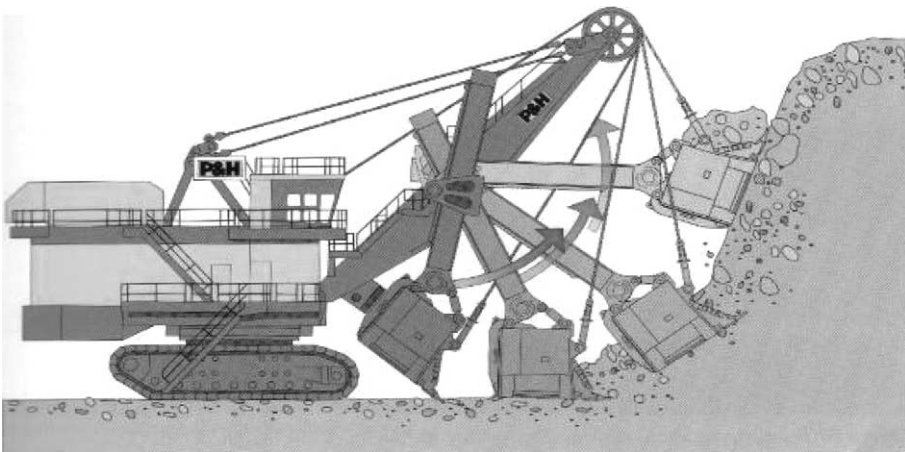


Fig. 3. The digging phase of a cable shovel [4].

efficient digging of excavated materials. Thus, a proper understanding of all motions involved in the digging cycle is essential for efficient and productive digging.

3. Dynamic modeling of the cable shovel

As discussed previously, the dynamic equations can be obtained by applying either Lagrange method or iterative Newton–Euler method. Here iterative Newton–Euler method is used because the iterative formulation lends itself to efficient software implementation, which is of importance for real-time simulation and parameter estimation [2,3]. Also, it provides detail information on all links and joints, which will be useful in possible stress and/or strength analysis of the components during excavation. The Newton–Euler dynamic algorithm for computing the crowd force and the hoist torque comprises two parts. First, the velocities and accelerations are iteratively computed from dipper handle to dipper and the Newton–Euler equations are applied to each of them. Second, the interactive forces and torques and joint actuator torques are computed recursively from dipper back to dipper handle [5,6].

The attention here is restricted to the boom, dipper handle and dipper assembly of the cable shovels because this research is only concerned with the digging section of a duty cycle. Therefore, the rotation of the upper structure is not modeled and the dipper handle and dipper assembly only moves on the vertical plane. The dynamic equation for cable shovel excavator is given as Eq. (1) and the procedure is summarized in the equations developed under [Appendix A](#).

$$D(\Theta)\ddot{\Theta} + C(\Theta, \dot{\Theta})\dot{\Theta} + G(\Theta) = F - F_{\text{load}}(F_t, F_n), \quad (1)$$

where $D(\Theta)$, $C(\Theta, \dot{\Theta})$ and $G(\Theta)$ are inertial matrix, Coriolis and centripetal effects, and gravity, respectively. Θ is a vector of generalized variables. F is related to the crowd and hoist force effects, and $F_{\text{load}}(F_t, F_n)$ is a resistive force due to soil-dipper interaction. For a cable shovel, the variables in equation can be summarized in Eqs. (2)–(5) as developed in [Appendix A](#).

$$D(\Theta) = \begin{bmatrix} m_1 + m_2 & -m_2 d_2 s_{2c_2} \\ -m_2 d_2 s_{2c_2} & I_{zz1} + I_{zz2} + m_1 d_1^2 + m_2 (l_1^2 + 2l_1 d_1 c_{2c_2} + d_1^2) \end{bmatrix}, \quad (2)$$

$$C(\Theta, \dot{\Theta}) = \begin{bmatrix} 0 & -(m_1 d_1 + m_2 (l_1 + d_2 c_{2c_2})) \dot{\theta}_1 \\ 2(m_1 d_1 + m_2 (l_1 + d_2 c_{2c_2})) \dot{\theta}_1 & 0 \end{bmatrix}, \quad (3)$$

$$G(\Theta) = \begin{bmatrix} (m_1 + m_2) g s_1 \\ (m_1 d_1 c_1 + m_2 (l_1 c_1 + d_2 c_{12c_2})) g \end{bmatrix}, \quad (4)$$

$$F_{\text{load}}(F_t, F_n) = \begin{bmatrix} F_t c_{2\theta_b} - F_n s_{2\theta_b} \\ F_t (l_1 + l_2) s_{2\theta_b} + F_n (l_1 + l_2) c_{2\theta_b} \end{bmatrix}. \quad (5)$$

It may be observed from Eq. (1) that F , which is the force provided by crowd and hoist motors, is mainly used to overcome two parts of external effects. One is the dynamic part, including the inertial effect, Coriolis and centripetal effects, and gravity. The other is the interaction force between the bucket and the surrounding media. The inertial part, $D(\theta)$, and gravity part, $G(\theta)$, summarize the effect of geometrical and material properties of the dipper handle and dipper, while $C(\theta, \dot{\theta})$ is an indicator of the kinematic and dynamic effect of the digging strategies including the digging profile and time distribution during the digging process. $F_{load}(F_t, F_n)$ is mainly related to the physical and mechanical properties of the materials being excavated and the geometrical characteristics of the dipper although other effects such as digging strategies may not be negligible. Other effects, which are not considered here, may include friction and losses in the transmission process. Therefore, from the dynamic model, it can be seen that the material properties to be excavated, the geometrical and physical properties of the dipper handle and dipper, as well as the digging strategies are the main factors in determining the required crowd and hoist forces, and thus, the performance of a cable shovel. The formulation of the dynamic model presented above depends on precise knowledge of the various kinematic and dynamic parameters, which characterize a particular cable shovel. These parameters can be estimated from data supplied by shovel manufacturers.

4. Numerical examples of cable shovel dynamics

A scaled cable shovel model is used here for the demonstration of the kinematics and dynamic behavior of the cable shovel during surface mining excavation. The main geometry and physical data for the shovel simulated is listed in Table 1. The digging trajectory shown in Fig. 4 is first generated based on the reference coordinate (x_w, y_w, z_w) in Fig. 10. Fig. 4 shows a generalized excavator bucket tip trajectory for completing the digging section of a working cycle using this reference coordinate. The bucket begins the digging cycle at point A and completes the cycle at point C. The excavator sits on the elevation A–B. The amount of digging below and/or above A–B elevation depends on the type of excavator. The bucket penetrates into formation at point A until it reaches a maximum

Table 1
Main shovel structural data for a scaled cable shovel

	Length (m)	Mass (kg)	Inertia moments ^a (kg m ²)
Handle	11.888 ^b	65800	7.749E + 5
Dipper	5.800 ^c	24500	6.458E + 4
Bucket capacity 30.6 m ³			

^a Moment of inertia about the gravitational center.
^b Length between the end of the handle and the joint for handle and bucket.
^c Length between the joint for handle and bucket and bucket tip.

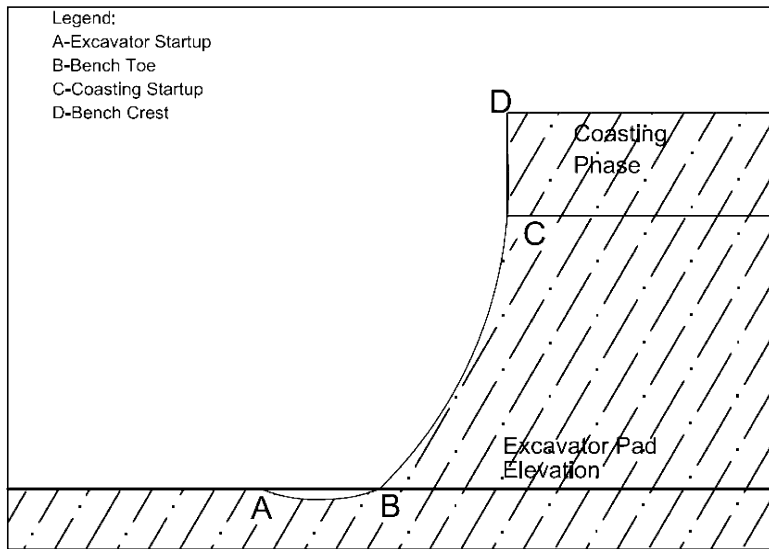


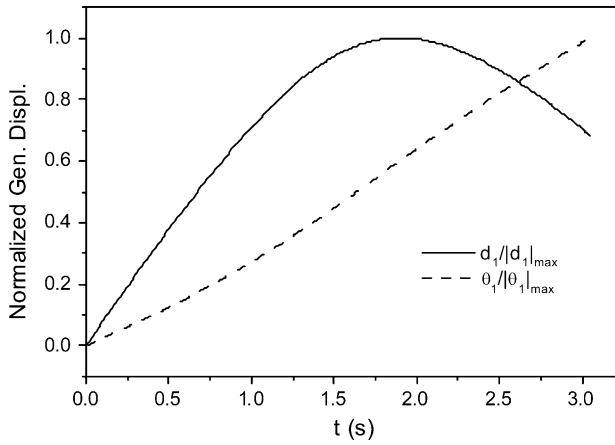
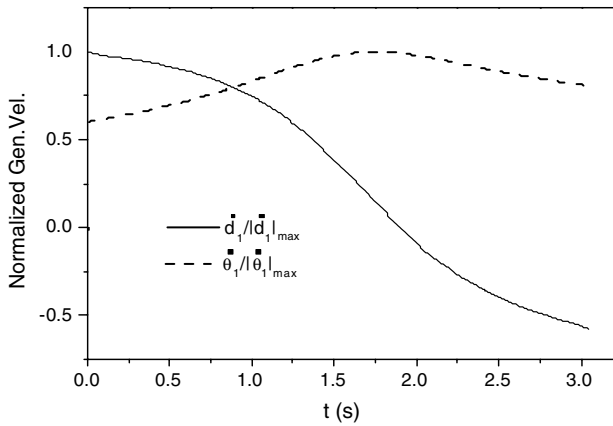
Fig. 4. A trajectory for a shovel dipper tip.

cut depth of 0.186 m. Then it starts to rise to be clear of the formation. It takes 3.5 s to finish the whole trajectory.

5. Discussions and analysis of results

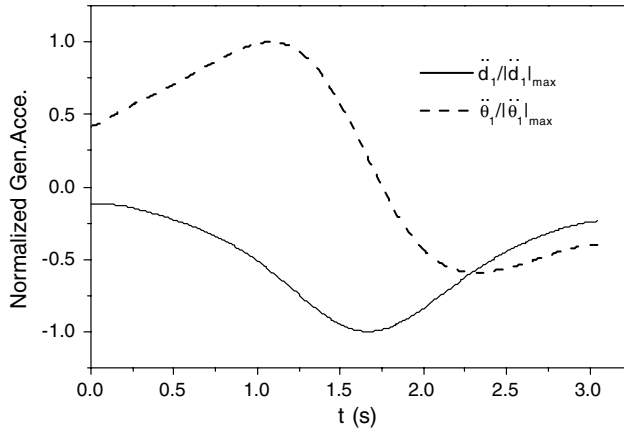
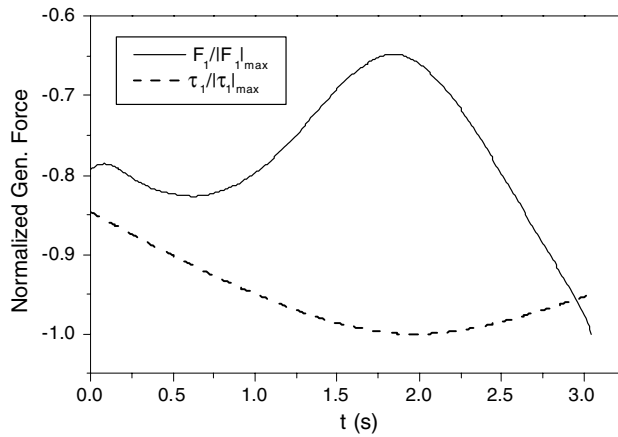
For the assumed dynamic trajectory applied to the dipper tip of the cable shovel as shown in Figs. 4–7 show the change of (d_1, θ_1) , $(\dot{d}_1, \dot{\theta}_1)$ and $(\ddot{d}_1, \ddot{\theta}_1)$ with time t , respectively. Note that in all these figures, the vertical axes have been normalized for a comparative purpose. In Fig. 5, the dipper handle, d_1 , advances at first, experiences a maximum displacement around the middle of the path profile ($t = 1.75$ – 2.0 s) and then retracts following the given digging profile, while θ_1 always increases with time. Although θ_1 is increasing constantly, its rate of change, $\dot{\theta}_1$, is not constant as illustrated in Fig. 6. It has a maximum around the path mid-point ($t = 1.75$ – 2.0 s).

For the dipper handle crowd rate, $\dot{d}_1$, it is decreasing constantly as shown in Fig. 6. The dipper handle crowd acceleration, $\ddot{d}_1$, and the angular acceleration, $\ddot{\theta}_1$, are more complex for the given digging path profile and are illustrated in Fig. 7. The angular acceleration first increases to a maximum at $t = 1.25$ s, and then decreases to a minimum at $t = 2.25$ s and gradually increases through the rest of the excavation cycle. The dipper handle crowd acceleration decreases at first, experiences a minimum acceleration at $t = 1.625$ s and then increases throughout the remaining excavation cycle. It is noted from these figures that the dipper handle crowd and angular rotation have different trends. The dipper handle crowd for the given digging trajectory has a much smaller change than that of the angular rotation, indicating a

Fig. 5. d_1 and θ_1 with time t .Fig. 6. $\dot{d}_1$ and $\dot{\theta}_1$ with time t .

predominant rotation effect in the inertial part. When $d_1, \dot{d}_1, \ddot{d}_1$, and $\theta_1, \dot{\theta}_1, \ddot{\theta}_1$ are determined for each time point, the left-hand side of Eqs. (A18) and (A19) can be calculated and their normalized changes with time are shown in Fig. 8. This figure shows that the dipper handle crowd inertial force and torque have opposite trends with time. The inertial force reaches a local maximum early in the cycle, gradually decreases to a local minimum, increases to a global maximum at $t = 1.875$ s and finally decreases sharply to a global minimum at the end of the cycle. The crowd arm torque decreases to a minimum at $t = 2.0$ s and gradually increases to the end of the cycle. It is also noted that there are some jaggednesses for $\dot{d}_1$ at the very early stage, which may be attributed to the time differential of the d_1 and θ_1 , not from numerical inaccuracies.

The cable shovel is designed with a breakout force, which is the maximum force available to overcome the resistive force from the material being excavated by the

Fig. 7. $\ddot{d}_1$ and $\ddot{\theta}_1$ with time t .Fig. 8. Inertial force and torque with time t .

shovel. The difference between the breakout force, F , and the resistive force, F_r , determines whether or not the formation can be excavated by the shovel. There are different models for predicting the resistive force, F_r . Zelenin et al. [7] provide a basis for estimating the resistive force F_r as shown in the following equation:

$$F_r = 10C_0d^{1.35}(1 + 2.6w)(1 + 0.0075\beta)(1 + 0.03s)e_zk_z, \quad (6)$$

where C_0 is compactness and cutting resistance index, d is depth of cutting, w is length of horizontal chip, β is the angle of cutting, s is the cutting edge index, e_z is the tool plate thickness and k_z is the index for the type of cutting. Based on the Zelenin model, the resistive force according to the current trajectory is calculated and shown in Fig. 9. The resistive force increases sharply from the beginning of the

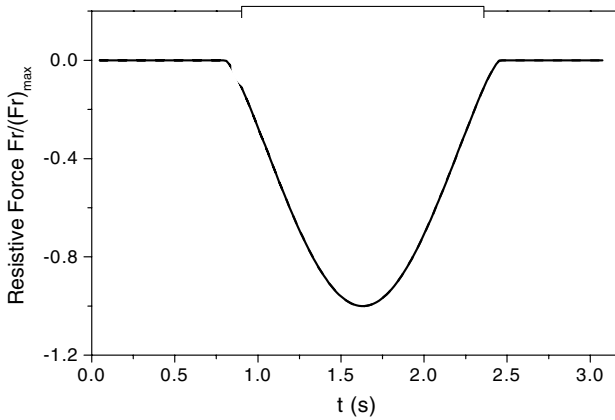


Fig. 9. Normalized resistive force with time.

excavation cycle to a maximum at $t = 1.625$ s and decreases from there at the same rate to the end of the excavation cycle. At $t = 1.625$ s, the breakout force of a given shovel is reached and further increase in the resistive force will require a shovel with greater breakout force or increased fragmentation. Alternatively, Eq. (1) may be solved for the breakout force, F , to select an appropriate shovel that can excavate the formation. Then the relationship between the crowd force and hoist torque from crowd and hoist motors and the resistive force F_r may be established. Therefore, by measuring the crowd force and hoist torque, one can know the current resistive force from the mined materials and, therefore, other information such as diggability.

6. Conclusions and contributions

Cable shovel dynamic models have been developed by analyzing the main functional components (the dipper handle and the dipper) as free body diagrams using an iterative Newton–Euler method. The models include the main factors that influence the performance of a cable shovel including the effect of both linear and angular motions of dipper handle and dipper. Based on the dynamic model, it is identified that the material properties to be excavated, the geometrical and physical properties of the dipper handle and dipper, as well as the digging strategies are the main factors that determine the required crowd and hoist forces and, thus, the performance of a cable shovel. The responses from a given digging trajectory are predicted as an illustration of the dynamic velocity and acceleration profiles. The kinematic results show that the critical phase occurs between 1.5 and 2.0 s of a 3-s excavation cycle with occurrence of maximum kinematic effects. The dynamic results also show a similar trend with maximum dynamic effects between 1.5 and 2.0 s. The results also show that the maximum resistive force occurs at 1.625 s within the excavation cycle. At this point the maximum breakout force of the equipment is reached and any increase

in the resistive load will require further fragmentation. The results provide appropriate information for excavation planning and execution. The next step in this research is the characterizing, modeling and simulation of the interaction between the shovel and the mining environments. This will eventually enable complete simulation of the shovel operation, as well as assessment of ore bench digging conditions based on on-line measurements of the information from the main functional components of the shovel excavators.

Appendix A. Detailed mathematical modeling of cable shovel dynamics

The digging path of the dipper is produced by the extension/retraction of the handle (crowd) and by the cable hosting action as in Fig. 3. Hoisting of the dipper is accomplished by means of cables attached to the dipper, which pass over sheaves at the boom point and spool on a deck mounted powered drum. The crowd action is produced either by cables or a direct rack and pinion gear drive. In Fig. 10, X_w, Y_w, Z_w are the coordinates at the lower and upper structures of the shovel, O_0, X_0, Y_0, Z_0 (base frame), O_1, X_1, Y_1, Z_1 and O_2, X_2, Y_2, Z_2 are the respective coordinates of the boom, crowd arm (dipper handle) and dipper. It is considered that the upper structures of the cable shovel are fixed, and thus, the kinematics and dynamic models of a cable shovel excavator are mainly related to dipper handle and dipper.

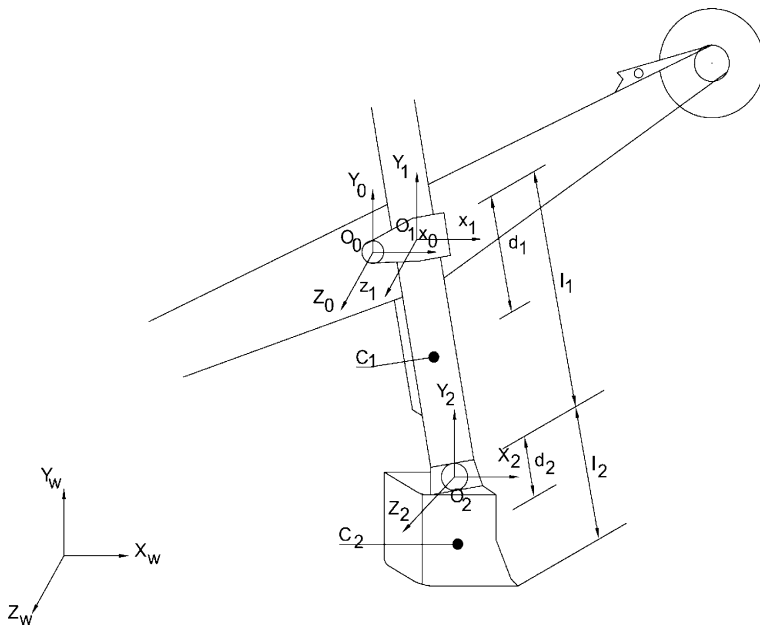


Fig. 10. Elements of shovel dynamics.

The Newton–Euler method is utilized to analyze the kinematics and dynamic behavior of the cable shovel. The advantage of this method is that it follows an iterative procedure and it is easy to implement in a simulation environment. Also, it provides detail information on all links and joints, which will be useful in possible stress and/or strength analysis of the components during excavation. The Newton–Euler dynamic algorithm for computing the crowd force and the hoist torque comprises two parts. First, the velocities and accelerations are iteratively computed from dipper handle to dipper and the Newton–Euler equations are applied to each of them. Second, the interactive forces and torques and joint actuator torques are computed recursively from dipper back to dipper handle [5]. The procedure is summarized in the equations below. The upper structure of the shovel is fixed, and hence it does not rotate during operation, resulting in the following equation:

$${}^0\omega_0 = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad \frac{d^0\omega_0}{dt} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad (\text{A1})$$

where ${}^0\omega_0$ and ${}^0\dot{\omega}_0 = (d^0\omega_0/dt)$ are the respective angular velocity and angular acceleration of the shovel boom. The first step of the Newton–Euler method is an outward iteration for the crowd arm. This procedure includes the calculation of the angular velocity, ${}^1\omega_1$, angular acceleration, ${}^1\dot{\omega}_1$, the linear acceleration at the gravity center of the crowd arm or dipper handle, ${}^1\dot{v}_{c_1}$, the torque at the gravity center of the crowd arm, 1N_1 , and the inertial force at the gravity center of the crowd arm, 1F_1 . The angular velocity is given by the equation

$${}^1\omega_1 = {}^0R^0\omega_0 + \dot{\theta}_1 {}^1\hat{Z}_1 = \begin{pmatrix} c_1 & s_1 & 0 \\ -s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{pmatrix}, \quad (\text{A2})$$

where 0R is the transformation matrix from base frame to the crowd arm based coordinate and $\dot{\theta}_1$ is the rotation angular velocity of the crowd arm based coordinate relative to the base frame. ${}^1\omega_1$ is necessary for estimating the Coriolis and centripetal effects of the crowd arm. The angular acceleration of the crowd arm is given by

$$\begin{aligned} {}^1\dot{\omega}_1 &= {}^0R^0\dot{\omega}_0 \times \dot{\theta}_1 {}^1\hat{Z}_1 + {}^0R^0\ddot{\omega}_0 + \ddot{\theta}_1 {}^1\hat{Z}_1 \\ &= \begin{pmatrix} c_1 & s_1 & 0 \\ -s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{pmatrix} + \begin{pmatrix} c_1 & s_1 & 0 \\ -s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \ddot{\theta}_1 \end{pmatrix} \\ &= \begin{pmatrix} 0 \\ 0 \\ \ddot{\theta}_1 \end{pmatrix}, \end{aligned} \quad (\text{A3})$$

where $\ddot{\theta}_1$ is the rotation angular acceleration of the crowd arm based coordinate relative to the base frame. ${}^1\dot{\omega}_1$ is used to calculate the angular moment of the crowd

arm. The linear acceleration, ${}^1\dot{v}_1$, at the connection point of the crowd arm is given by the equation

$$\begin{aligned} {}^1\dot{v}_1 &= {}^0R({}^0\dot{\omega}_0 \times {}^0P_1 + {}^0\omega_0 \times ({}^0\omega_0 \times {}^0P_1) + {}^0\dot{v}_0) + 2{}^1\omega_1 \times \dot{d}_1 {}^1\hat{X}_1 + \ddot{d}_1 {}^1\hat{X}_1 \\ &= \begin{pmatrix} gs_1 + \ddot{d}_1 \\ gc_1 + 2\dot{d}_1\dot{\theta}_1 \\ 0 \end{pmatrix}, \end{aligned} \quad (\text{A4})$$

where 0P_1 is a position vector, which locates the rotation point of the crowd arm. $\dot{d}_1$ and $\ddot{d}_1$ are the linear velocity and linear acceleration of the crowd arm, respectively. The gravity effect ${}^0\dot{v}_0$ is calculated as ${}^0\dot{v}_0 = g\hat{y}_0$. The linear acceleration at the gravity center of the crowd arm is given by

$${}^1\dot{v}_{c_1} = {}^1\dot{\omega}_1 \times {}^1P_{c_1} + {}^1\omega_1 \times ({}^1\omega_1 \times {}^1P_{c_1}) + {}^1\dot{v}_1 = \begin{pmatrix} -d_1\dot{\theta}_1^2 + \ddot{d}_1 + gs_1 \\ d_1\ddot{\theta}_1 + 2\dot{d}_1\dot{\theta}_1 + gc_1 \\ 0 \end{pmatrix}, \quad (\text{A5})$$

where ${}^1P_{c_1}$ is the position vector, which locates the center of the gravity of the crowd arm. ${}^1\dot{v}_{c_1}$ is used to calculate the inertial force of the crowd arm. Euler's equation is used to calculate the torque, 1N_1 , at the gravity center of the crowd arm in the equation

$$\begin{aligned} {}^1N_1 &= {}^{c_1}I_1 {}^1\dot{\omega}_1 + {}^1\omega_1 \times {}^{c_1}I_1 {}^1\omega_1 \\ &= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & I_{zz1} \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \ddot{\theta}_1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{pmatrix} \times \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & I_{zz1} \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{pmatrix} \right) \\ &= \begin{pmatrix} 0 \\ 0 \\ I_{zz1}\ddot{\theta}_1 \end{pmatrix}. \end{aligned} \quad (\text{A6})$$

Newton's equation is applied to calculate the inertial force, 1F_1 , at the gravity center of the crowd arm in the equation

$${}^1F_1 = m_1 {}^1\dot{v}_{c_1} = \begin{pmatrix} -m_1 d_1 \dot{\theta}_1^2 + m_1 \ddot{d}_1 + m_1 gs_2 \\ m_1 d_1 \ddot{\theta}_1 + 2m_1 \dot{d}_1 \dot{\theta}_1 + m_1 gc_1 \\ 0 \end{pmatrix}. \quad (\text{A7})$$

The same procedure can be applied to the dipper to calculate its angular velocity, ${}^2\omega_2$; angular acceleration, ${}^2\dot{\omega}_2$; the linear acceleration at the gravity center of the dipper, ${}^2\dot{v}_{c_2}$; the torque at the gravity center of the dipper, 2N_2 ; and the inertial force at the gravity center of the dipper, 2F_2 . They are

$${}^2\omega_2 = {}^2R {}^1\omega_1 + \dot{\theta}_2 {}^2\hat{Z}_2 = \begin{pmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{pmatrix}, \quad (\text{A8})$$

$$\begin{aligned}
{}^2\dot{\omega}_2 &= {}^2R^1\dot{\omega}_1 \times \dot{\theta}_2 {}^2\hat{Z}_2 + {}^2R^1\dot{\omega}_1 + \ddot{\theta}_2 {}^2\hat{Z}_2 \\
&= \begin{pmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{pmatrix} \times \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} c_2 & s_2 & 0 \\ -s_2 & c_2 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \ddot{\theta}_1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \\
&= \begin{pmatrix} 0 \\ 0 \\ \ddot{\theta}_1 \end{pmatrix}, \tag{A9}
\end{aligned}$$

$$\begin{aligned}
{}^2\dot{v}_2 &= {}^2R({}^1\dot{\omega}_1 \times {}^1P_2 + {}^1\omega_1 \times ({}^1\omega_1 \times {}^1P_2) + {}^1\dot{v}_1) + 2{}^2\omega_2 \times \dot{d}_2 {}^2\hat{X}_2 + \ddot{d}_2 {}^2\hat{X}_2 \\
&= \begin{pmatrix} -l_1\dot{\theta}_1^2 c_2 + \ddot{d}_1 c_2 + l_1\ddot{\theta}_1 s_2 + 2\dot{d}_1\dot{\theta}_1 s_2 + g s_{12} \\ l_1\dot{\theta}_1^2 s_2 - \ddot{d}_1 s_2 + l_1\ddot{\theta}_1 c_2 + 2\dot{d}_1\dot{\theta}_1 c_2 + g c_{12} \\ 0 \end{pmatrix}, \tag{A10}
\end{aligned}$$

$$\begin{aligned}
{}^2\dot{v}_{c_2} &= {}^2\dot{\omega}_2 \times {}^2P_{c_2} + {}^2\omega_2 \times ({}^2\omega_2 \times {}^2P_{c_2}) + {}^2\dot{v}_2 \\
&= \begin{pmatrix} -(l_1 c_2 + d_2 c_{c_2})\dot{\theta}_1^2 + (l_1 s_2 - d_2 s_{c_2})\ddot{\theta}_1 + 2\dot{d}_1\dot{\theta}_1 s_2 + \ddot{d}_1 c_2 + g s_{12} \\ (l_1 s_2 - d_2 s_{c_2})\dot{\theta}_1^2 + (l_1 c_2 + d_2 c_{c_2})\ddot{\theta}_1 + 2\dot{d}_1\dot{\theta}_1 c_2 - \ddot{d}_1 s_2 + g c_{12} \\ 0 \end{pmatrix}, \tag{A11}
\end{aligned}$$

$$\begin{aligned}
{}^2N_2 &= {}^cI_2 {}^2\dot{\omega}_2 + {}^2\omega_2 \times {}^cI_2 {}^2\omega_2 \\
&= \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & I_{zz2} \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \ddot{\theta}_1 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{pmatrix} \times \left(\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & I_{zz4} \end{pmatrix} \begin{pmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{pmatrix} \right) \\
&= \begin{pmatrix} 0 \\ 0 \\ I_{zz2}\ddot{\theta}_1 \end{pmatrix}, \tag{A12}
\end{aligned}$$

$$\begin{aligned}
{}^2F_2 &= m_2 {}^2\dot{v}_{c_2} \\
&= \begin{pmatrix} -m_2(l_1 c_2 + d_2 c_{c_2})\dot{\theta}_1^2 + m_2(l_1 s_2 - d_2 s_{c_2})\ddot{\theta}_1 + 2m_2\dot{d}_1\dot{\theta}_1 s_2 + m_2\ddot{d}_1 c_2 + m_2 g s_{12} \\ m_2(l_1 s_2 - d_2 s_{c_2})\dot{\theta}_1^2 + m_2(l_1 c_2 + d_2 c_{c_2})\ddot{\theta}_1 + 2m_2\dot{d}_1\dot{\theta}_1 c_2 - m_2\ddot{d}_1 s_2 + m_2 g c_{12} \\ 0 \end{pmatrix}. \tag{A13}
\end{aligned}$$

Having computed the forces and torques acting on the crowd arm and the dipper, it now remains to calculate the joint torques which will result in these net forces and torques being applied to the crowd arm and the dipper. This is achieved by writing force and moment balance equations based on free body diagrams of the crowd arm and the dipper. The inward iteration for the dipper is first calculated as follows:

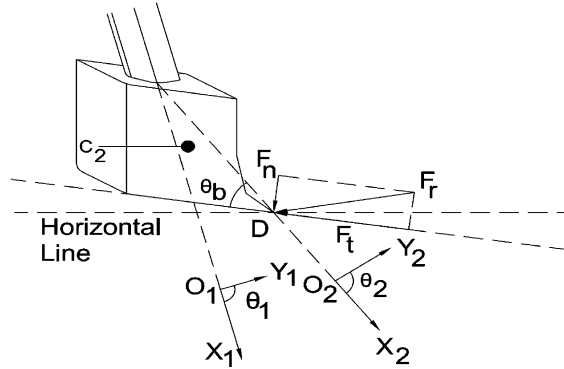


Fig. 11. Interaction between dipper and environment.

$$\begin{aligned}
 {}^2f_2 &= {}^3R^3 f_3 + {}^2F_2 \\
 &= \begin{pmatrix} -m_2(l_1 c_2 + d_2 c_{c_2}) \ddot{\theta}_1^2 + m_2(l_1 s_2 - d_2 s_{c_2}) \ddot{\theta}_1 + 2m_2 \dot{d}_1 \dot{\theta}_1 s_2 + m_2 \ddot{d}_1 c_2 + F_t c_{\theta_b} - F_n s_{\theta_b} + m_2 g s_{12} \\ m_2(l_1 s_2 - d_2 s_{c_2}) \dot{\theta}_1^2 + m_2(l_1 c_2 + d_2 c_{c_2}) \ddot{\theta}_1 + 2m_2 \dot{d}_1 \dot{\theta}_1 c_2 - m_2 \ddot{d}_1 s_2 + F_t s_{\theta_b} + F_n c_{\theta_b} + m_2 g c_{12} \\ 0 \end{pmatrix},
 \end{aligned} \quad (A14)$$

where F_t and F_n are the tangential and normal components of the interaction force, respectively, between the dipper tip and the media (Fig. 11). Eq. (A14) is the force balance equation for the dipper. The torque balance equation for the dipper is given by the equation

$$\begin{aligned}
 {}^2n_2 &= {}^2N_2 + {}^3R^3 n_3 + {}^2P_{c_2} \times {}^2F_2 + {}^2P_3 \times {}^3R^3 f_3 \\
 &= \begin{pmatrix} 0 \\ 0 \\ I_{zz2} \ddot{\theta}_1 + m_2 l_1 d_2 s_2 c_2 \dot{\theta}_1^2 + m_2(l_1 c_2 c_2 + d_2) d_2 \ddot{\theta}_1 + 2m_2 d_2 \dot{d}_1 \dot{\theta}_1 c_2 c_2 - m_2 d_2 \ddot{d}_1 s_2 c_2 \end{pmatrix} \\
 &\quad + \begin{pmatrix} 0 \\ 0 \\ m_2 d_2 g c_{12 c_2} + F_t l_2 s_2 \theta_b + F_n l_2 c_2 \theta_b \end{pmatrix}.
 \end{aligned} \quad (A15)$$

Following the same procedure, the inward iteration for the crowd arm is also given by

$$\begin{aligned}
 {}^1f_1 &= {}^2R^2 f_2 + {}^1F_1 \\
 &= \begin{pmatrix} -(m_1 d_1 + m_2(l_1 + d_2 c_2 c_2)) \ddot{\theta}_1^2 - m_2 d_2 s_2 c_2 \ddot{\theta}_1 + (m_1 + m_2) \ddot{d}_1 + F_t c_2 \theta_b - F_n s_2 \theta_b + (m_1 + m_2) g s_1 \\ -m_2 d_2 s_2 c_2 \dot{\theta}_1^2 + (m_1 d_1 + m_2(l_1 + d_2 c_2 c_2)) \ddot{\theta}_1 + 2(m_1 + m_2) \dot{d}_1 \dot{\theta}_1 + F_t s_2 \theta_b + F_n c_2 \theta_b + (m_1 + m_2) g c_1 \\ 0 \end{pmatrix},
 \end{aligned} \quad (A16)$$

$$\begin{aligned}
{}^1n_1 = {}^1N_1 + {}^1_2R^2n_2 + {}^1P_{c_1} \times {}^1F_1 + {}^1P_2 \times {}^1_2R^2f_2 = & \begin{pmatrix} 0 \\ 0 \\ (I_{zz1} + I_{zz2})\ddot{\theta}_1 \end{pmatrix} \\
& + \begin{pmatrix} 0 \\ 0 \\ (m_1d_1^2 + m_2(l_1^2 + 2l_1d_1c_{2c_2} + d_1^2))\ddot{\theta}_1 + 2(m_1d_1 + m_2(l_1 + d_2c_{2c_2}))\dot{d}_1\dot{\theta}_1 - m_2d_2\ddot{d}_1s_{2c_2} \end{pmatrix} \\
& + \begin{pmatrix} 0 \\ 0 \\ (m_1d_1c_1 + m_2(l_1c_1 + d_2c_{12c_2}))g \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ F_t(l_1 + l_2)s_{2\theta_b} + F_n(l_1 + l_2)c_{2\theta_b} \end{pmatrix}. \quad (A17)
\end{aligned}$$

Eqs. (A16) and (A17) are the respective force and moment balance equations for the crowd arm. Finally, the equations for the crowd inertial force and torque are obtained in the following equations:

$$\begin{aligned}
F_1 = (m_1 + m_2)\ddot{d}_1 - (m_1d_1 + m_2(l_1 + d_2c_{2c_2}))\dot{\theta}_1^2 \\
- m_2d_2s_{2c_2}\ddot{\theta}_1 + (m_1 + m_2)gs_1 + F_tc_{2\theta_b} - F_ns_{2\theta_b}, \quad (A18)
\end{aligned}$$

$$\begin{aligned}
\tau_1 = (I_{zz1} + I_{zz2})\ddot{\theta}_1 + (m_1d_1^2 + m_2(l_1^2 + 2l_1d_1c_{2c_2} + d_1^2))\ddot{\theta}_1 \\
+ 2(m_1d_1 + m_2(l_1 + d_2c_{2c_2}))\dot{d}_1\dot{\theta}_1 - m_2d_2\ddot{d}_1s_{2c_2} \\
+ (m_1d_1c_1 + m_2(l_1c_1 + d_2c_{12c_2}))g + F_t(l_1 + l_2)s_{2\theta_b} + F_n(l_1 + l_2)c_{2\theta_b}. \quad (A19)
\end{aligned}$$

Eqs. (A18) and (A19) can be rearranged concisely as follows:

$$\begin{aligned}
\begin{Bmatrix} F_1 \\ \tau_1 \end{Bmatrix} = & \begin{bmatrix} m_1 + m_2 & -m_2d_2s_{2c_2} \\ -m_2d_2s_{2c_2}I_{zz1} + I_{zz2} + m_1d_1^2 + m_2(l_1^2 + 2l_1d_1c_{2c_2} + d_1^2) \end{bmatrix} \begin{Bmatrix} \ddot{d}_1 \\ \ddot{\theta}_1 \end{Bmatrix} \\
& + \begin{bmatrix} 0 & -(m_1d_1 + m_2(l_1 + d_2c_{2c_2}))\dot{\theta}_1 \\ 2(m_1d_1 + m_2(l_1 + d_2c_{2c_2}))\dot{\theta}_1 & 0 \end{bmatrix} \begin{Bmatrix} \dot{d}_1 \\ \dot{\theta}_1 \end{Bmatrix} \\
& + \begin{bmatrix} (m_1 + m_2)gs_1 \\ (m_1d_1c_1 + m_2(l_1c_1 + d_2c_{12c_2}))g \end{bmatrix} \begin{Bmatrix} d_1 \\ \theta_1 \end{Bmatrix} \\
& + \begin{bmatrix} F_tc_{2\theta_b} - F_ns_{2\theta_b} \\ F_t(l_1 + l_2)s_{2\theta_b} + F_n(l_1 + l_2)c_{2\theta_b} \end{bmatrix}. \quad (A20)
\end{aligned}$$

If Θ is assumed to be $\{d_1 \ \theta_1\}^T$ for cable shovels, $D(\Theta)$, $C(\Theta, \dot{\Theta})$, $G(\Theta)$ and $F_{load}(F_t, F_n)$ in Eqs. (2)–(5) can be determined from Eqs. (A20) and (A20) becomes the dynamic model for the motion of cable shovels illustrated by Eq. (1).

References

- [1] Frimpong S, Szymanski J, Pedrycz W, Gao Y. Intelligent shovel excavation in varying oil sands formation and bitumen content. A proposal submitted to COURSE, University of Alberta, Edmonton, Canada; 2001.

- [2] Hendricks C, Daneshmend L, Wu S, Scoble M. Design of a simulator for productivity analysis of electric mining shovels. In: Proceedings of 2nd international symposium on mine mechanization and Automation, Lulea, Sweden; 1993.
- [3] Daneshmend L, Hendricks C, Wu S, Scoble M. Design of a mining shovel simulator. Innovative mine design for the 21st century, Kingston, Ont., Canada; 1993.
- [4] Peak performance: excavator selection. P & H MinePro Services; 2003.
- [5] Craig JJ. Introduction to robotics: mechanics and control. Reading (MA): Addison-Wesley; 1986.
- [6] Vaha PK, Skibniewski MJ. Dynamic model of excavator. J Aerospace Eng 1993;6:148–66.
- [7] Zelenin AN, Balovnev VI, Kerov IP. Machines for moving the earth. New Delhi: Amerind Publishing; 1985.